Blueberry Diseases

Identification, causes, favourable conditions, management and verified remedies

Seven major diseases of an emerging Indian crop, drawn mainly from US land-grant sources (Southern highbush focus)

Primary sources: LSU AgCenter (Southern highbush), UConn, Michigan State, Oregon State / PNW Handbook, University of Maine and UNH extension; ICAR Indian Horticulture (2020) for the Indian context. No CIB&RC blueberry claim was identified, so no doses are given. Compiled 18 October 2026.

Contents

About this document

Scope and method

Each disease chapter follows the same six headings: identification; causes and progression; favourable conditions; prevention and management; remedies and treatment; and sources. Disease descriptions come mainly from US land-grant university extension publications and the ICAR Indian Horticulture article on blueberry. No fungicide or bactericide claim for blueberry could be matched to the CIB&RC compilation, and no Indian university package for blueberry diseases was found, so chemical information is shown only as active ingredients with US efficacy ratings (tag US), without doses, for reference.

Verification tags used in the chemical tables

US - US reference only. The active ingredient and efficacy rating (E excellent, VG very good, G good, F fair, P poor; R = resistance possible) come from the LSU AgCenter guide for US-labelled products. NO dose is given, because no CIB&RC label claim on blueberry was identified in this review. These are not Indian recommendations.

-- - Not recommended / no chemical option verified. The sources reviewed do not support a chemical recommendation, so none is given.

Important cautions

- No CIB&RC-registered fungicide or bactericide claim for blueberry could be identified in this review (no blueberry entry was found in the CIB&RC fungicide compilation up to 30/09/2025 - single products pp. 2-41 and combinations pp. 42-79 - read from a copy on a university website; the live register and the insecticide compilation were not checked for blueberry). Doses are therefore deliberately not given. Before any chemical use in India, check the live CIB&RC register (cibrc.gov.in) and the product label for blueberry, and consult your KVK or state agricultural university.
- Blueberry disease science here is drawn mainly from US land-grant universities (LSU AgCenter for Southern highbush, plus UConn, Michigan State, Oregon State and others). Weather thresholds and calendar timings refer to US conditions and must be adapted to local climate and growth stage.
- Blueberry is an emerging crop in India. ICAR (Indian Horticulture, 2020) states commercial production had not been reported, with trials at CSKHPKV Palampur (Himachal Pradesh) since 2007-08 (ICH); trade press reports growers in Maharashtra, Karnataka, Himachal Pradesh, Uttarakhand, Madhya Pradesh and Uttar Pradesh testing low-chill Southern highbush varieties, often in high tunnels and soilless media, and notes the lack of standardised Indian protocols (FRP). No ICAR / SAU disease package for blueberry was found.
- Soil and water: blueberry needs acidic soil (ideal pH 4.2-4.8 per ICAR; ICH) and well-drained, aerated media. Many Indian disease problems will be linked to waterlogging and root health.

Summary table

Disease	Pathogen / type	Main non-chemical measures	Chemical status
Phytophthora Root Rot	Phytophthora cinnamomi (Oomycete (soil-borne))	Well-drained site or raised beds, clean plants, avoid waterlogging	US reference only (no dose); chemicals do not rescue severely infected plants
Botrytis Blight (Gray Mould)	Botrytis cinerea (Fungal (necrotrophic; sclerotia, wind-borne spores))	Airflow, pruning, sanitation, drip irrigation	US reference only (no dose)
Anthracnose and Ripe Rot (Fruit Rots)	Colletotrichum gloeosporioides (also Alternaria and Phomopsis fruit rots) (Fungal (twig- and debris-borne))	Frequent picking, quick cooling, airflow, avoid excess nitrogen	US reference only (no dose)
Phomopsis Twig Blight and Botryosphaeria Stem Blight / Canker	Phomopsis vaccinii; Botryosphaeria spp. (Fungal (stem-borne))	Avoid wounds and drought stress, remove diseased canes, dormant lime sulphur	US reference only; limited for Botryosphaeria
Mummy Berry	Monilinia vaccinii-corymbosi (Fungal (sclerotia-borne, two infection phases))	Remove or bury mummies, resistant cultivars; relevance to India unconfirmed	US reference only (no dose)
Leaf and Fruit Spots (Septoria, Exobasidium, Rust)	Septoria spp.; Exobasidium maculosum; rust fungi (Fungal (leaf-debris and bud-scale-borne))	Airflow, debris management, bud-swell lime sulphur	US reference only (no dose)
Bacterial Leaf Scorch	Xylella fastidiosa (Bacterial (insect-vectored, xylem-limited))	Clean plants, resistant varieties, vector control	None (no chemicals available)

Quick separation of look-alikes: yellowing, stunting and dead fine roots in wet soil point to Phytophthora root rot; gray mould on blossoms, twigs and fruit points to Botrytis; salmon-coloured spore masses on ripe fruit point to anthracnose; dying year-old stems or cankers point to Phomopsis/Botryosphaeria; cream-pink fruit that shrivel into hard mummies point to mummy berry; purple-bordered light leaf spots or orange rust pustules point to the leaf spots; marginal leaf browning with twig yellowing in Southern highbush points to bacterial leaf scorch (or a root problem, so check roots).

1. Phytophthora Root Rot

Blueberry | Phytophthora cinnamomi | Oomycete (soil-borne)

At a glance. One of the most significant blueberry diseases. Kills fine roots in wet, poorly drained soil, worst on young plants. Prevention through drainage, raised beds and clean planting stock matters more than any chemical.

1. Identification

Symptoms

- Leaves turn yellow and plant growth stops; rootlets are killed and infected roots and crowns are discoloured; bushes may defoliate and die (LSU).
- Plants become stunted and off-colour with much reduced berry production; an entire plant can die within one growing season (UCT).
- Symptoms are usually most severe on young plants; leaf-margin scorch can develop as roots are compromised; older plants with larger root systems may survive for years with chronic infection (PNR).

Distinguishing features

- Marginal leaf browning can also be caused by bacterial leaf scorch (LSU), so examine the roots and crown before deciding.
- Armillaria root rot is slow, shows black 'shoestring' rhizomorphs or mushrooms at the base, and follows clearing of woodland (UCT).

2. Causes and progression

- Pathogen: Phytophthora cinnamomi (LSU, PNR). It survives in soil as chlamydospores (LSU).
- Zoospores swim towards fine roots, infect and kill the root system (PNR).
- Progression: loss of fine roots reduces water and nutrient uptake, leading to yellowing, stunting, scorch, wilting and death (LSU, PNR).

3. Favourable conditions

- Wet soil conditions and temperatures between about 68 and 90 F (20-32 C) (LSU).
- Root rot is much more severe in poorly drained, heavy soils (PNR); the disease is nearly always associated with poorly drained low spots (UCT).
- ICAR notes that waterlogged soil is detrimental and can cause root infection by rot pathogens such as Phytophthora and Pythium (ICH).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Plant only disease-free plants, in well-drained soil or on raised beds (LSU, PNR, ICH).
- Avoid heavy clay soils and low spots that puddle; improve drainage before planting (UCT).
- Raised beds are advised in areas of heavy rainfall combined with heavy soils (ICH).
- Higher phosphorus and potassium appear to help by stimulating root production (PNR).
- Cultivar Patriot is the only one reported to show resistance (UCT, US data).

Regional note (India)

Raised beds and soilless media (cocopeat, perlite) are used by Indian growers to manage the acidic, well-drained root zone that blueberry needs (FRP; trade press). No Indian incidence data were found.

5. Remedies and treatment

Chemical options with verification status

Active ingredient	Use / target	US-guide efficacy (no India dose)	Water	Tag	Basis
Mefenoxam (soil, while plants are dormant)	Phytophthora root rot	Rated G	n/a	US	LSU guide (US label)
Oxathiapiprolin (soil, dormant)	Phytophthora root rot	Rated G	n/a	US	LSU guide (US label)
Phosphite (mono-/di-potassium phosphite; foliar)	Phytophthora root rot	Rated VG	n/a	US	LSU guide (US label)

Active ingredient	Use / target	US-guide efficacy (no India dose)	Water	Tag	Basis
Aluminium tris (fosetyl-Al; foliar)	Phytophthora root rot	Rated G	n/a	US	LSU guide (US label)

Even in the US guide, chemicals are not likely to restore healthy growth in severely infected plants (LSU), and the Oregon State handbook says chemical control is not recommended for plants showing moderate to severe symptoms and that these fungicide groups do not kill the organism but can delay symptoms (PNR). No CIB&RC blueberry claim was found, so no dose is given.

6. Sources and citations

[LSU] LSU AgCenter (2025). Louisiana Plant Disease Management Guide - Blueberry, Publication 1802 (revised Oct 2024; covers Southern highbush and rabbiteye). https://www.lsuagcenter.com/~/media/system/2/e/4/0/2e40f4d1dd1e8dce7b732afd272a8838/p1802_2025-louisianaplantdiseaseblueberrypdf.pdf

[PNR] Pacific Northwest Plant Disease Management Handbook (Oregon State University et al.): Blueberry - Root rot (Phytophthora cinnamomi). https://pnwhandbooks.org/node/2318

[UCT] University of Connecticut IPM: Blueberry disease management (reviewed 2012; conditions of the US Northeast). https://ipm.cahnr.uconn.edu/blueberry-disease-management

[ICH] Negi N.D. & Upadhyay S.K. (2020). Blueberry: a future super fruit. Indian Horticulture, Nov-Dec 2020, ICAR-DKMA (CSKHPKV Palampur). https://epubs.icar.org.in/index.php/IndHort/article/download/111443/43679/289748

[FRP] FreshPlaza / Food Business Africa (trade press): India's blueberry growers adapt with high tunnels, low-chill varieties and soilless production (supplementary, secondary source). https://www.freshplaza.com/europe/article/9838851/india-s-blueberry-growers-adapt-with-high-tunnels-low-chill-varieties-and-soilless-production/

2. Botrytis Blight (Gray Mould)

Blueberry | Botrytis cinerea | Fungal (necrotrophic; sclerotia, wind-borne spores)

At a glance. Gray mould of blossoms, twigs and fruit, after cool, wet weather and on frost-injured blossoms. Airflow and sanitation are the base; fruit infected early may only rot after harvest.

1. Identification

Symptoms

- The fungus attacks tender young twigs, leaves, flowers and fruit, turning them brown to black; infected fruit do not rot until after harvest (LSU).
- Infected tissue typically has a gray cast from mycelium and spores (UCT).
- In favourable conditions it can cause stem blight (UCT).

Distinguishing features

- Stem symptoms are indistinguishable from Phomopsis twig blight; the fungus must be cultured to confirm it (UCT).
- Blueberry scorch (a virus disease) also causes blossom blight, but it lacks the gray mould appearance that Botrytis shows (UCT).

2. Causes and progression

- Pathogen: Botrytis cinerea; survives as dormant mycelium or sclerotia (LSU) on dead twigs, prunings, fruit left on the bush and as a saprophyte on dead organic matter (PNB).
- Spores are wind-dispersed (LSU). It is a weak pathogen that tends to infect blossoms injured by light frost (UCT).

3. Favourable conditions

- Outbreaks are infrequent and usually follow cool, rainy weather in May (a US spring) (UCT).
- Tight fruit clusters (Blueray, Rancocas, Weymouth) are especially susceptible (UCT).
- The sources reviewed give no Indian temperature/humidity thresholds.

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Plant in areas with good air movement and prune for open canopies (LSU, UCT).
- Remove dead twigs, prunings and left-over fruit that carry sclerotia (PNB).
- Prefer drip irrigation; if using overhead irrigation, apply early in the morning so leaves, flowers and berries dry quickly (UCT).

Regional note (India)

No Indian incidence data were found. Protected cultivation can raise humidity, so airflow and sanitation are important (general; see PNB, LSU).

5. Remedies and treatment

Chemical options with verification status

Active ingredient	Use / target	US-guide efficacy (no India dose)	Water	Tag	Basis
Fenhexamid	Botrytis blight	Rated E	n/a	US	LSU guide (US label)
Cyprodinil + fludioxonil	Botrytis blight	Rated E	n/a	US	LSU guide (US label)
Pyraclostrobin + boscalid	Botrytis blight	Rated E	n/a	US	LSU guide (US label)
Captan	Botrytis blight	Rated F	n/a	US	LSU guide (US label)
Ziram	Botrytis blight	Rated F	n/a	US	LSU guide (US label)
Fluazinam	Botrytis blight	Rated F	n/a	US	LSU guide (US label)

In the US guide, sprays target bloom (10-20 % bloom to full bloom) and are used only if needed (LSU); UConn advises starting at mid-bloom if favourable weather persists, repeating 7-10 days later through petal fall (UCT). These are US schedules, shown for information only. No CIB&RC blueberry claim was found, so no dose is given.

6. Sources and citations

[LSU] LSU AgCenter (2025). Louisiana Plant Disease Management Guide - Blueberry, Publication 1802 (revised Oct 2024; covers Southern highbush and rabbiteye). https://www.lsuagcenter.com/~/media/system/2/e/4/0/2e40f4d1dd1e8dce7b732afd272a8838/p1802_2025-louisianaplantdiseaseblueberrypdf.pdf

[UCT] University of Connecticut IPM: Blueberry disease management (reviewed 2012; conditions of the US Northeast). https://ipm.cahnr.uconn.edu/blueberry-disease-management

[PNB] Pacific Northwest Plant Disease Management Handbook: Blueberry disease descriptions (Botrytis blight, cankers, scorch). https://pnwhandbooks.org/handbook/plant-disease?page=203

[MSB] Michigan State University Extension: Blueberry scouting guide (mummy berry; apothecia and infection temperature). https://www.canr.msu.edu/blueberries/uploads/files/BlueberryGuide-online-FINAL.pdf

3. Anthracnose and Ripe Rot (Fruit Rots)

Blueberry | Colletotrichum gloeosporioides (also Alternaria and Phomopsis fruit rots) | Fungal (twig- and debris-borne)

At a glance. Fruit rots that develop as berries ripen, often showing salmon-coloured spore masses. Frequent picking, quick cooling and air circulation reduce losses; this is a major post-harvest quality issue.

1. Identification

Symptoms

- A variety of rots develop on the fruit as they mature (LSU).
- Anthracnose: infected berries have a distinctive area of salmon-coloured spores on the surface; it can also cause lesions on stems and leaves (UCT).

Distinguishing features

- The salmon-coloured spore mass separates anthracnose from other fruit rots (UCT).
- Mummy berry fruit shrivel into hard mummies (LSU); Botrytis fruit show gray mould (UCT).

2. Causes and progression

- Fungi causing fruit rots survive from year to year on or in infected twigs; Alternaria also survives in fallen debris (LSU).
- Anthracnose spreads from infected to uninfected berries during favourable weather, especially if berries are picked wet (UCT).

3. Favourable conditions

- UConn notes spread is rapid in 'favourable weather' and that heavy nitrogen fertilisation tends to induce higher disease incidence (UCT).
- Berkeley, Bluecrop, Blueray, Coville and Jersey are highly susceptible (UCT, US data).
- No Indian temperature/humidity thresholds were found.

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Harvest ripe fruit regularly and cool it soon after harvest to minimise post-harvest disease (LSU).
- Pick berries frequently, but not when wet (UCT).
- Ensure good air circulation around fruit clusters; avoid excess nitrogen (UCT).

Regional note (India)

No Indian incidence data were found. Fruit quality and shelf life are the main concern for Indian fresh-market growers (general; LSU post-harvest advice).

5. Remedies and treatment

Chemical options with verification status

Active ingredient	Use / target	US-guide efficacy (no India dose)	Water	Tag	Basis
Metconazole	Ripe rot / anthracnose fruit rot	Rated E	n/a	US	LSU guide (US label)
Cyprodinil + fludioxonil	Ripe rot / anthracnose fruit rot	Rated E	n/a	US	LSU guide (US label)
Pydiflumetofen + fludioxonil	Ripe rot / anthracnose fruit rot	Rated E	n/a	US	LSU guide (US label)
Pyraclostrobin + boscalid	Ripe rot (resistance possible)	Rated E, R	n/a	US	LSU guide (US label)
Azoxystrobin	Ripe rot (resistance possible)	Rated E, R	n/a	US	LSU guide (US label)
Captan	Ripe rot / anthracnose fruit	Rated G	n/a	US	LSU guide (US label)

Active ingredient	Use / target	US-guide efficacy (no India dose)	Water	Tag	Basis
	rot				
Fluazinam	Ripe rot / anthracnose fruit rot	Rated G	n/a	US	LSU guide (US label)

In the US guide, sprays start at bloom and continue through petal fall, cover sprays at 7-14 day intervals, and pre-harvest/harvest sprays for fruit rots (LSU); UConn says anthracnose sprays start at full bloom at 7-10 day intervals (UCT). Shown for information only; no CIB&RC blueberry claim was found, so no dose is given. Residue limits (MRL) and PHI for export or domestic markets must be checked on the Indian label.

6. Sources and citations

[LSU] LSU AgCenter (2025). Louisiana Plant Disease Management Guide - Blueberry, Publication 1802 (revised Oct 2024; covers Southern highbush and rabbiteye). https://www.lsuagcenter.com/~/media/system/2/e/4/0/2e40f4d1dd1e8dce7b732afd272a8838/p1802_2025-louisianaplantdiseaseblueberrypdf.pdf

[UCT] University of Connecticut IPM: Blueberry disease management (reviewed 2012; conditions of the US Northeast). https://ipm.cahnr.uconn.edu/blueberry-disease-management

4. Phomopsis Twig Blight and Botryosphaeria Stem Blight / Canker

Blueberry | Phomopsis vaccinii; Botryosphaeria spp. | Fungal (stem-borne)

At a glance. Dieback of canes and branches, often after cold, drought or mechanical injury. Sanitation, avoiding wounds and drought stress, and removing diseased wood are the main controls; fungicides are limited.

1. Identification

Symptoms

- Phomopsis twig blight: flower-bearing year-old stems die (LSU). Small cankers may occur on stems, or whole branches may be killed when lesions coalesce; crown infection can kill the plant (UCT).
- Botryosphaeria canker: lesions appear on new growth and turn into cankers on susceptible varieties. Botryosphaeria stem blight: individual branches die (LSU).

Distinguishing features

- Botrytis stem blight is indistinguishable from Phomopsis without culturing (UCT).
- Phomopsis can also cause fruit rot, which is uncommon in New England (UCT).

2. Causes and progression

- The pathogens survive from year to year in infected stems; spores are spread by wind or rain; Phomopsis infects stems through flowers (LSU).
- Incidence increases when plants have been stressed by cold or drought injury (UCT).

3. Favourable conditions

- Plant stress (cold, drought) and wounds from pruning or cultivation favour infection (UCT).
- The sources reviewed give no Indian weather thresholds.

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Avoid mechanical injury to stems; remove diseased canes and shoots (LSU).
- Avoid injury from improper pruning or cultivating, and irrigate to avoid drought stress (UCT).
- A dormant lime-sulphur treatment is described as the most important fungicide application for Phomopsis twig blight; apply in late winter or early spring before buds break dormancy and avoid once growth has started, or young tissue will be injured (UCT).
- Discourage late-season growth and encourage early hardening off through proper fertilisation (UCT).
- Cultivars that are more resistant to Phomopsis: Bluejay, Bluetta, Cabot, Darrow, Duke, Elliott, Jersey, Pemberton, Pioneer, Rancocas, Rubel and Stanley; highly susceptible: Berkeley, Bluecrop, Blueray, Coville, Weymouth (UCT; US data).

Regional note (India)

No Indian incidence data were found. Pruning-wound care and water management are the practical controls (UCT, LSU).

5. Remedies and treatment

Chemical options with verification status

Active ingredient	Use / target	US-guide efficacy (no India dose)	Water	Tag	Basis
Lime sulphur (calcium polysulfide; dormant season)	Phomopsis twig blight	Described by UConn as the most important application; not rated for Phomopsis in LSU table	n/a	US	UConn (UCT)
Fenbuconazole	Phomopsis twig blight	Rated E	n/a	US	LSU guide (US label)
Metconazole	Phomopsis twig blight	Rated E	n/a	US	LSU guide (US label)
Pyraclostrobin + boscalid	Phomopsis twig blight	Rated E	n/a	US	LSU guide (US label)
Ziram	Phomopsis twig	Rated G	n/a	US	LSU guide (US label)

Active ingredient	Use / target	US-guide efficacy (no India dose)	Water	Tag	Basis
	blight				

Fungicides are not likely to be very effective for Botryosphaeria (LSU), so sanitation and avoiding wounds are the main controls for that pathogen. The rows above are US-guide information for Phomopsis only; no CIB&RC blueberry claim was found, so no dose is given.

6. Sources and citations

[LSU] LSU AgCenter (2025). Louisiana Plant Disease Management Guide - Blueberry, Publication 1802 (revised Oct 2024; covers Southern highbush and rabbiteye). https://www.lsuagcenter.com/~/media/system/2/e/4/0/2e40f4d1dd1e8dce7b732afd272a8838/p1802_2025-louisianaplantdiseaseblueberrypdf.pdf

[UCT] University of Connecticut IPM: Blueberry disease management (reviewed 2012; conditions of the US Northeast). https://ipm.cahnr.uconn.edu/blueberry-disease-management

[PNB] Pacific Northwest Plant Disease Management Handbook: Blueberry disease descriptions (Botrytis blight, cankers, scorch). https://pnwhandbooks.org/handbook/plant-disease?page=203

5. Mummy Berry

Blueberry | Monilinia vaccinii-corymbosi | Fungal (sclerotia-borne, two infection phases)

At a glance. A cool-spring disease of highbush and lowbush blueberry that turns fruit into hard mummies. It needs overwintering mummies and cool, wet weather, so its relevance to low-chill Indian plantings is unconfirmed.

1. Identification

Symptoms

- The fungus attacks and kills leaves, twigs, flowers and fruit; infected fruit turn cream to pink as they begin to mature, then shrivel and harden into 'mummies' (LSU).
- Primary phase: leaf and shoot blight from overwintered ascospores infecting emerging buds; secondary phase: conidia from infected tissue infect blossoms and young berries (UCT).
- Infected berries are tan or pink; hardened berries shrivel and fall to the ground (UCT).

Distinguishing features

- Mummified, shrivelled fruit are characteristic; there is zero tolerance for mummified fruit in the market (MSM).
- Blossom blight can resemble Botrytis, but Botrytis shows gray mould (UCT).

2. Causes and progression

- Pathogen: Monilinia vaccinii-corymbosi (UCT, LSU).
- Mummified fruit (sclerotia) survive winter in fallen leaf litter and germinate in spring if the soil stays wet long enough, producing goblet-shaped apothecia; ascospores are released for about a month from bud break (UME).
- Secondary conidia are carried by wind, rain and bees to blossoms (UNH).

3. Favourable conditions

- Cool, rainy weather in May (a US spring) favours infection (UCT).
- Optimum temperature for apothecia formation and infection: 50-57 F (10-14 C) (MSB).
- At temperatures in the 60s F a wet or foggy period of at least one day is needed for infection (UME).
- Because this life cycle depends on overwintering mummies and cool, wet springs, its risk in low-chill Indian plantings in warm regions is unconfirmed; no Indian report was found.

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Remove or bury mummified fruit to 1 inch depth (LSU); rake and discard mummies in small plantings, bury them by cultivation or 2 inches of mulch in larger ones (UCT).
- Resistance to shoot-blight phase: Bluejay, Bluecrop, Darrow, Duke, Elliott, Jersey, Stanley; susceptible: Bluehaven, Bluegold, Coville, Northblue, Sierra (UCT; US data).
- Avoid equipment that has been in orchards infected with mummy berry (LSU).
- Pruning for airflow and drip irrigation help manage foliar fungal diseases (UCT).

Regional note (India)

No Indian report of mummy berry was found. Because the disease depends on a cold winter and cool spring, it is more plausible in Himachal Pradesh-type climates than in warm-region protected cultivation, but this is unconfirmed.

5. Remedies and treatment

Chemical options with verification status

Active ingredient	Use / target	US-guide efficacy (no India dose)	Water	Tag	Basis
Fenbuconazole	Mummy berry	Rated E	n/a	US	LSU guide (US label)
Metconazole	Mummy berry	Rated E	n/a	US	LSU guide (US label)
Propiconazole	Mummy berry	Rated E	n/a	US	LSU guide (US label)
Prothioconazole	Mummy berry	Rated E	n/a	US	LSU guide (US label)
Pyraclostrobin + boscalid	Mummy berry	Rated VG	n/a	US	LSU guide (US label)

Active ingredient	Use / target	US-guide efficacy (no India dose)	Water	Tag	Basis
Captan	Mummy berry	Rated F	n/a	US	LSU guide (US label)

The US guide stresses that the first sprays of the season matter most for mummy berry, starting at green tip or when 1-5 % of blooms are open (LSU); UConn targets primary infection with a first spray at green tip and a second 10 days later (UCT). These are US schedules, shown for information only; no CIB&RC blueberry claim was found, so no dose is given.

6. Sources and citations

[LSU] LSU AgCenter (2025). Louisiana Plant Disease Management Guide - Blueberry, Publication 1802 (revised Oct 2024; covers Southern highbush and rabbiteye). https://www.lsuagcenter.com/~/media/system/2/e/4/0/2e40f4d1dd1e8dce7b732afd272a8838/p1802_2025-louisianaplantdiseaseblueberrypdf.pdf

[UCT] University of Connecticut IPM: Blueberry disease management (reviewed 2012; conditions of the US Northeast). https://ipm.cahnr.uconn.edu/blueberry-disease-management

[MSB] Michigan State University Extension: Blueberry scouting guide (mummy berry; apothecia and infection temperature). https://www.canr.msu.edu/blueberries/uploads/files/BlueberryGuide-online-FINAL.pdf

[MSM] Michigan State University Extension Bulletin E-2846: Mummy berry facts. https://www.canr.msu.edu/blueberries/uploads/files/E2846 Mummy Berry Facts.pdf

[UME] University of Maine Cooperative Extension Fact Sheet 211: Blueberry diseases (mummy berry). https://extension.umaine.edu/blueberries/?p=1223

[UNH] University of New Hampshire Extension: Mummy berry. https://extension.unh.edu/resource/mummy-berry

6. Leaf and Fruit Spots (Septoria, Exobasidium, Rust)

Blueberry | Septoria spp.; Exobasidium maculosum; rust fungi | Fungal (leaf-debris and bud-scale-borne)

At a glance. Spotting of leaves (and, for Exobasidium, fruit) that can cause defoliation and downgrade fruit. Pruning for airflow, debris management and well-timed dormant-season sprays are the US recommendations.

1. Identification

Symptoms

- Septoria: small, circular, light-coloured lesions with a purple border on leaves; sunken lesions may form on stems (LSU).
- Rust: spots on upper leaf sides change from yellow to red to brown, with orange spores underneath (LSU).
- Exobasidium fruit and leaf spot: whitish spots on leaf undersides with light green spots above, turning brown; fruit spots stay unripe while the rest ripens (LSU).

Distinguishing features

- Septoria has a purple-bordered, light-centred spot; rust has orange spore masses on the underside; Exobasidium spots appear on both leaf sides and on fruit (LSU).

2. Causes and progression

- Septoria survives in infected leaf debris and stems; spores are windborne (LSU).
- Rust is believed to survive on evergreen Vaccinium plants (LSU).
- Exobasidium is thought to overwinter on plant surfaces, bud scales or bark and infect in spring; its life cycle is still being studied (LSU).

3. Favourable conditions

- The sources reviewed give no thresholds; good air movement and canopy penetration are the recommended controls (LSU).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Plant in areas with good air movement; prune to allow air movement and spray penetration (LSU).
- Manage leaf debris, which carries Septoria between seasons (LSU).
- Lime sulphur applied at bud swell, 1-2 weeks before leaf or flower buds break, has been found very effective against Exobasidium (LSU).

Regional note (India)

No Indian incidence data were found.

5. Remedies and treatment

Chemical options with verification status

Active ingredient	Use / target	US-guide efficacy (no India dose)	Water	Tag	Basis
Lime sulphur (calcium polysulfide; bud swell)	Exobasidium	Rated E	n/a	US	LSU guide (US label)
Captan	Exobasidium	Rated VG	n/a	US	LSU guide (US label)
Metconazole	Septoria leaf spot	Rated E	n/a	US	LSU guide (US label)
Fenbuconazole	Septoria leaf spot	Rated E	n/a	US	LSU guide (US label)
Phosphite (foliar)	Septoria leaf spot	Rated VG	n/a	US	LSU guide (US label)
Copper products labelled for blueberry	Rust	Rated F	n/a	US	LSU guide (US label)

The US guide recommends post-harvest sprays for leaf spots in summer and fall and notes sprays are needed only if the disease is confirmed (LSU). Shown for information only; no CIB&RC blueberry claim was found, so no dose is given.

6. Sources and citations

[LSU] LSU AgCenter (2025). Louisiana Plant Disease Management Guide - Blueberry, Publication 1802 (revised Oct 2024; covers Southern highbush and rabbiteye). https://www.lsuagcenter.com/~/media/system/2/e/4/0/2e40f4d1dd1e8dce7b732afd272a8838/p1802_2025-louisianaplantdiseaseblueberrypdf.pdf

7. Bacterial Leaf Scorch

Blueberry | Xylella fastidiosa | Bacterial (insect-vectored, xylem-limited)

At a glance. A bacterial scorch of Southern highbush that spreads through infected propagation material and sharpshooter leafhoppers. There is no chemical cure; clean plants, resistant varieties and vector control are the tools.

1. Identification

Symptoms

- In susceptible Southern highbush varieties, leaf edges turn brown (marginal necrosis) and leaves eventually drop; young twigs and stems may turn yellow before plants die (LSU).
- Rabbiteye cultivars may show early fall colour and a gradual decline in fruit production (LSU).

Distinguishing features

- Leaf-margin scorch also accompanies Phytophthora root rot (PNR), so check roots; blueberry scorch virus (a different disease) is pollen-borne (PNB).

2. Causes and progression

- Pathogen: Xylella fastidiosa (LSU).
- Infected plants serve as reservoirs; the bacterium is transmitted through propagation and by sharpshooter insects (LSU).

3. Favourable conditions

- The sources reviewed give no weather thresholds for this disease.

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Plant disease-free plants and choose resistant varieties when planting Southern highbush blueberries (LSU).
- Control leafhopper (sharpshooter) vectors; the LSU guide refers to the Southeast Regional Blueberry Integrated Management Guide for insecticide recommendations (LSU).

Regional note (India)

Southern highbush varieties are the type being tested in India (FRP; trade press), so certified clean planting material is especially important. No Indian incidence data were found.

5. Remedies and treatment

Chemical options with verification status

No chemical recommendation is given (tag --).

No chemicals are available for managing this disease (LSU). No vector-insecticide dose is given here, because none could be verified against the CIB&RC register for blueberry.

6. Sources and citations

[LSU] LSU AgCenter (2025). Louisiana Plant Disease Management Guide - Blueberry, Publication 1802 (revised Oct 2024; covers Southern highbush and rabbiteye). https://www.lsuagcenter.com/~/media/system/2/e/4/0/2e40f4d1dd1e8dce7b732afd272a8838/p1802_2025-louisianaplantdiseaseblueberrypdf.pdf

[PNB] Pacific Northwest Plant Disease Management Handbook: Blueberry disease descriptions (Botrytis blight, cankers, scorch). https://pnwhandbooks.org/handbook/plant-disease?page=203

[PNR] Pacific Northwest Plant Disease Management Handbook (Oregon State University et al.): Blueberry - Root rot (Phytophthora cinnamomi). https://pnwhandbooks.org/node/2318

[FRP] FreshPlaza / Food Business Africa (trade press): India's blueberry growers adapt with high tunnels, low-chill varieties and soilless production (supplementary, secondary source). https://www.freshplaza.com/europe/article/9838851/india-s-blueberry-growers-adapt-with-high-tunnels-low-chill-varieties-and-soilless-production/

Consolidated source list

[LSU] LSU AgCenter (2025). Louisiana Plant Disease Management Guide - Blueberry, Publication 1802 (revised Oct 2024; covers Southern highbush and rabbiteye). https://www.lsuagcenter.com/~/media/system/2/e/4/0/2e40f4d1dd1e8dce7b732afd272a8838/p1802_2025-louisianaplantdiseaseblueberrypdf.pdf

[PNR] Pacific Northwest Plant Disease Management Handbook (Oregon State University et al.): Blueberry - Root rot (Phytophthora cinnamomi). https://pnwhandbooks.org/node/2318

[UCT] University of Connecticut IPM: Blueberry disease management (reviewed 2012; conditions of the US Northeast). https://ipm.cahnr.uconn.edu/blueberry-disease-management

[ICH] Negi N.D. & Upadhyay S.K. (2020). Blueberry: a future super fruit. Indian Horticulture, Nov-Dec 2020, ICAR-DKMA (CSKHPKV Palampur). https://epubs.icar.org.in/index.php/IndHort/article/download/111443/43679/289748

[FRP] FreshPlaza / Food Business Africa (trade press): India's blueberry growers adapt with high tunnels, low-chill varieties and soilless production (supplementary, secondary source). https://www.freshplaza.com/europe/article/9838851/india-s-blueberry-growers-adapt-with-high-tunnels-low-chill-varieties-and-soilless-production/

[PNB] Pacific Northwest Plant Disease Management Handbook: Blueberry disease descriptions (Botrytis blight, cankers, scorch). https://pnwhandbooks.org/handbook/plant-disease?page=203

[MSB] Michigan State University Extension: Blueberry scouting guide (mummy berry; apothecia and infection temperature). https://www.canr.msu.edu/blueberries/uploads/files/BlueberryGuide-online-FINAL.pdf

[MSM] Michigan State University Extension Bulletin E-2846: Mummy berry facts. https://www.canr.msu.edu/blueberries/uploads/files/E2846 Mummy Berry Facts.pdf

[UME] University of Maine Cooperative Extension Fact Sheet 211: Blueberry diseases (mummy berry). https://extension.umaine.edu/blueberries/?p=1223

[UNH] University of New Hampshire Extension: Mummy berry. https://extension.unh.edu/resource/mummy-berry

Items checked but not included

- Indian disease package: no ICAR, state agricultural university or KVK package of practices for blueberry diseases was found; disease information is from US land-grant universities.
- CIB&RC: no blueberry entry was found in the CIB&RC fungicide compilation up to 30/09/2025 (single and combination pages); the live register and the insecticide compilation were not checked for blueberry. Doses are therefore not given.
- Not covered: Armillaria root rot, blueberry scorch / shock / mosaic viruses, powdery mildew, crown gall, nematodes and insect pests.
- Indian incidence: no field survey data for these diseases on blueberry in India were found.